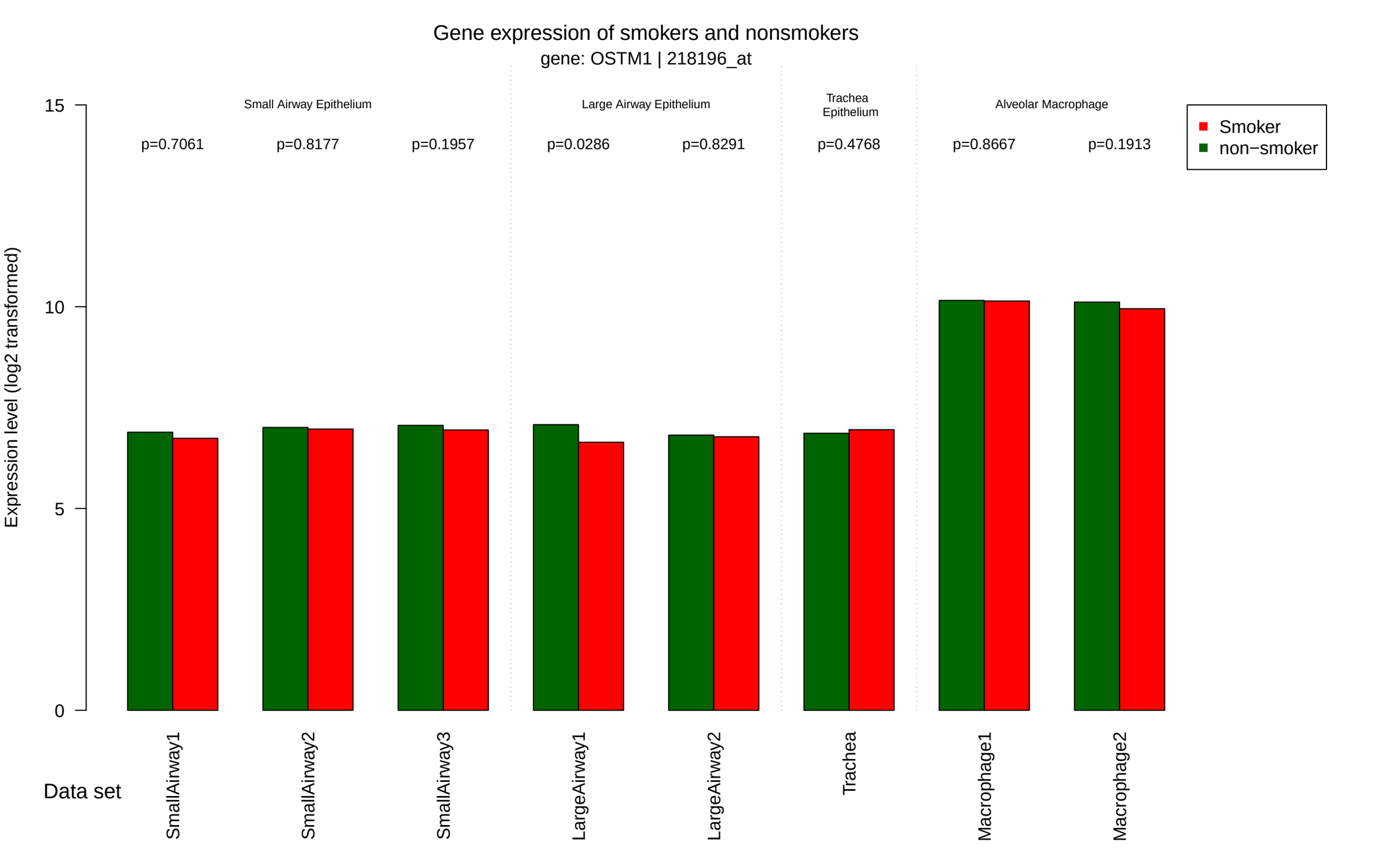
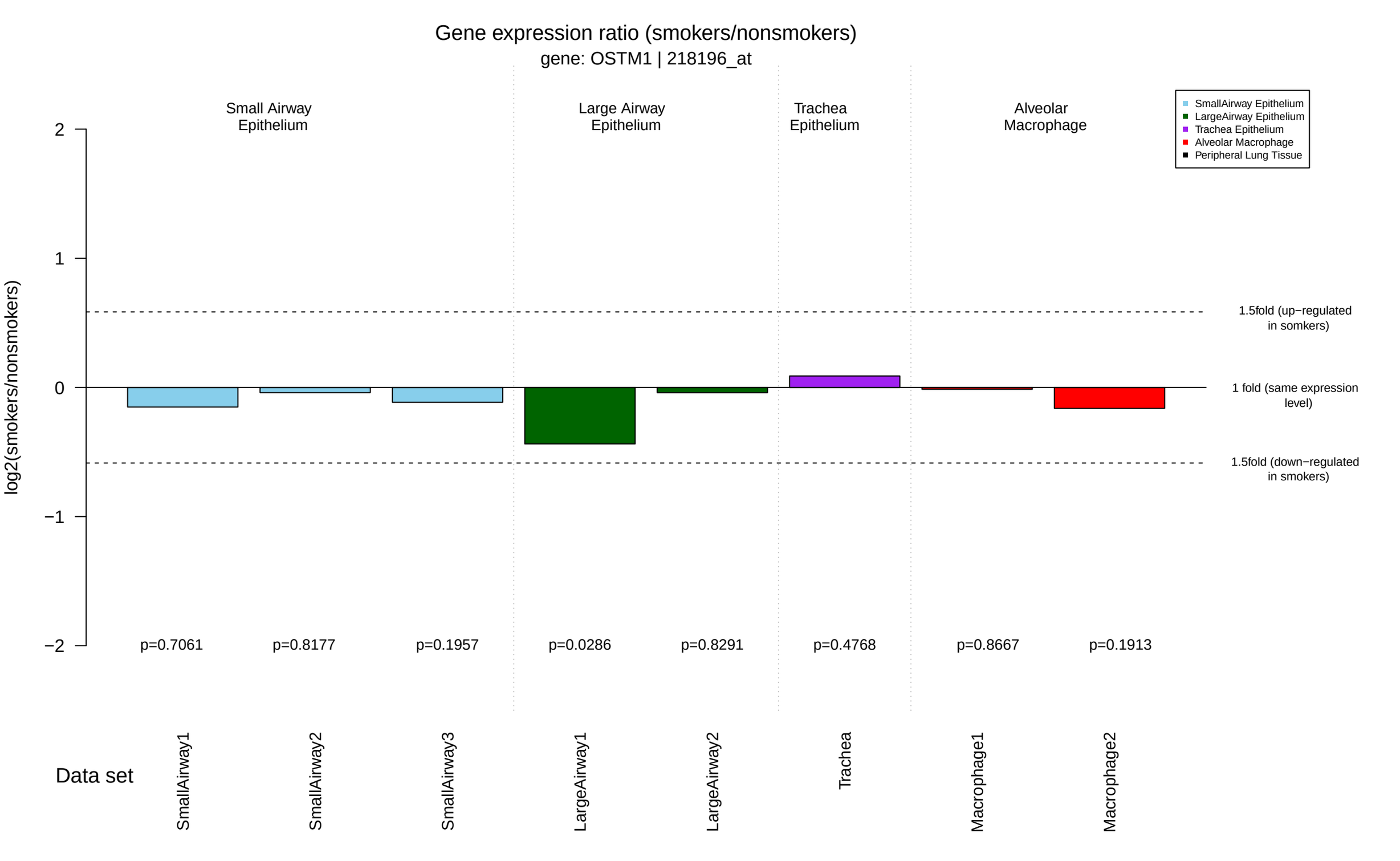
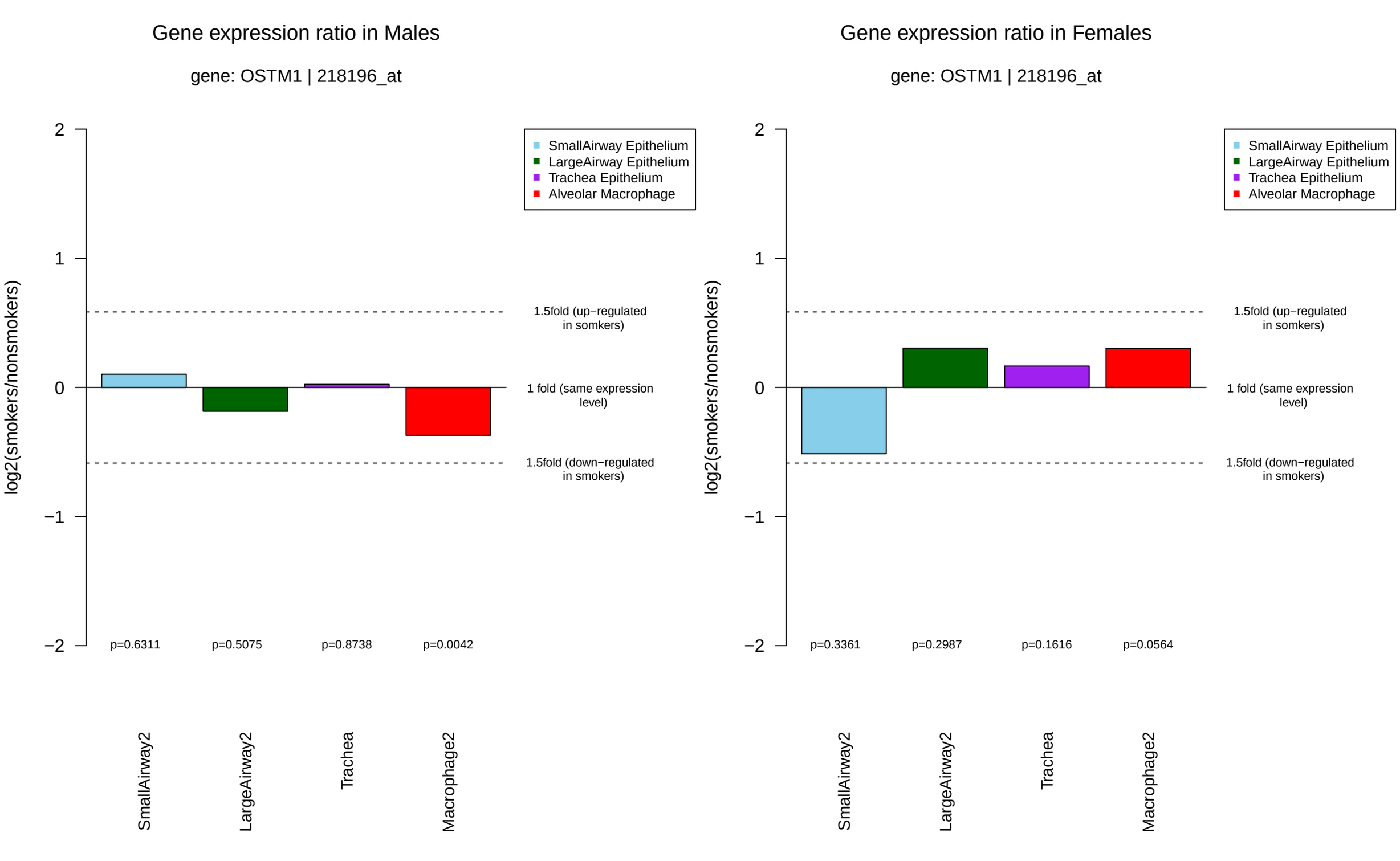
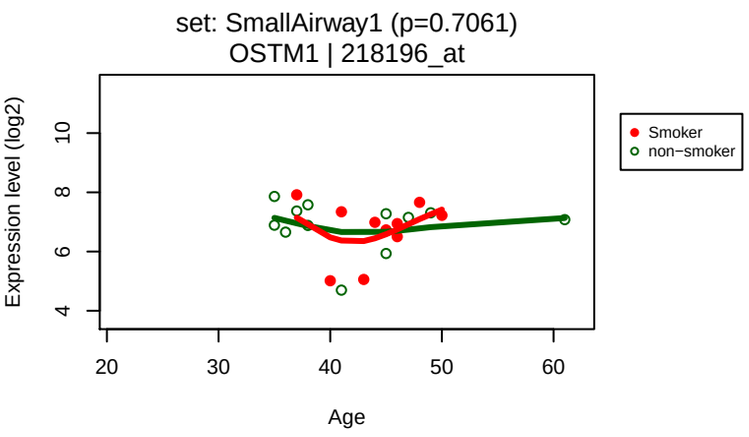
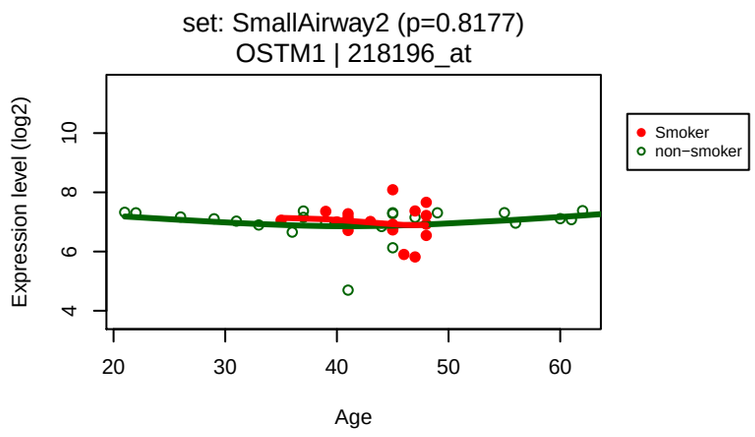
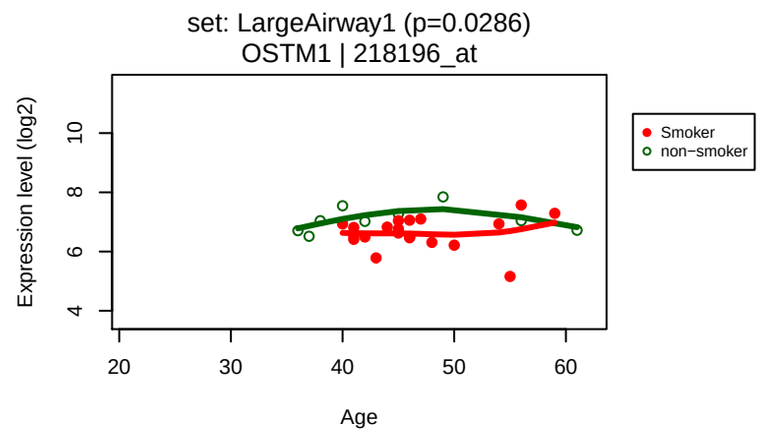

Gene expression of smokers and nonsmokers
gene: OSTM1 | 218196_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7061
p=0.8177
p=0.1957
p=0.0286
p=0.8291
p=0.4768
p=0.8667
p=0.1913
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

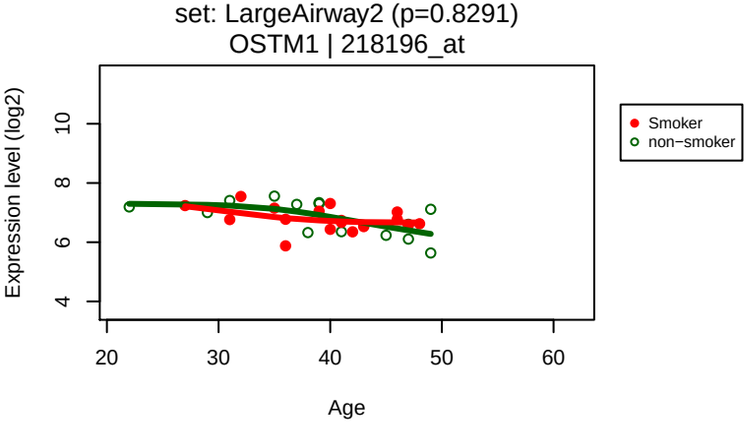

Gene expression ratio (smokers/nonsmokers)
gene: OSTM1 | 218196_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7061
p=0.8177
p=0.1957
p=0.0286
p=0.8291
p=0.4768
p=0.8667
p=0.1913
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

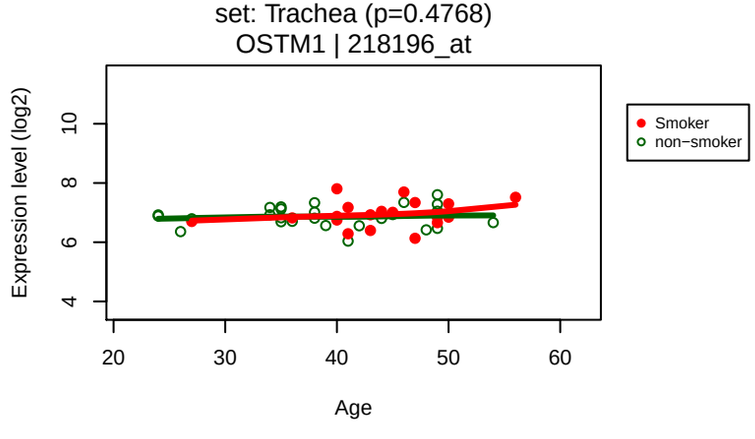

Gene expression ratio in Males
gene: OSTM1 | 218196_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6311
p=0.5075
p=0.8738
p=0.0042
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OSTM1 | 218196_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3361
p=0.2987
p=0.1616
p=0.0564
SmallAirway2
LargeAirway2
Trachea
Macrophage2

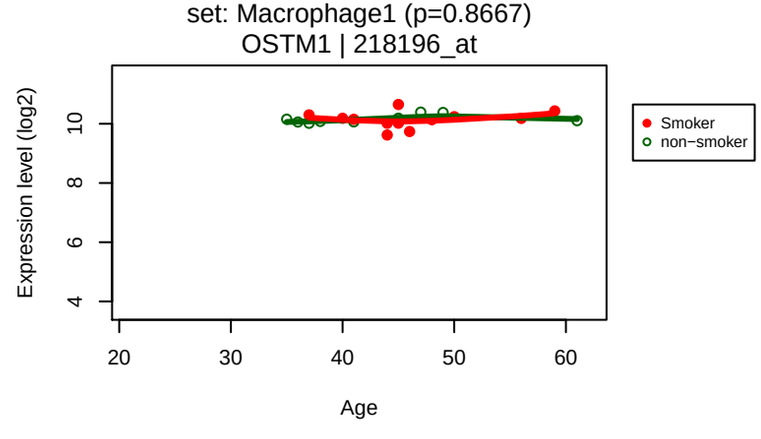

set: SmallAirway1 (p=0.7061)
OSTM1 | 218196_at
Expression level (log2)
Age
Smoker
non−smoker

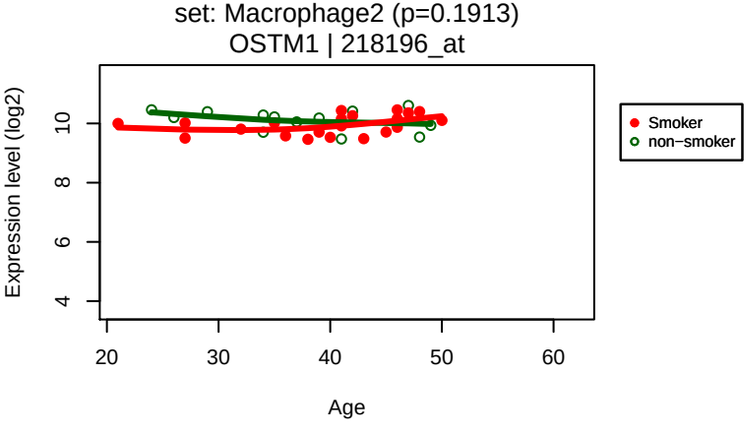

set: SmallAirway2 (p=0.8177)
OSTM1 | 218196_at
Expression level (log2)
Age
Smoker
non−smoker

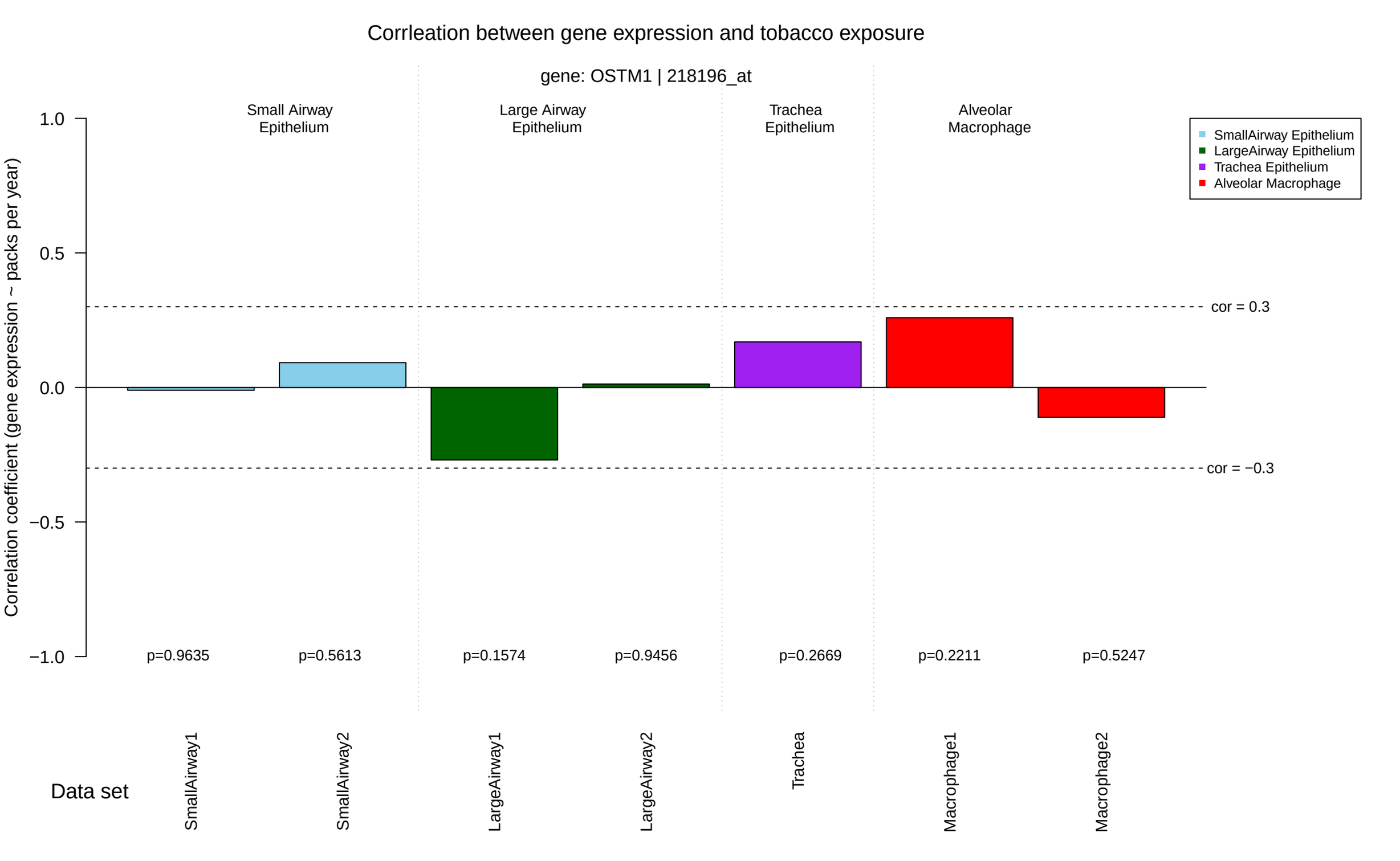

set: LargeAirway1 (p=0.0286)
OSTM1 | 218196_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8291)
OSTM1 | 218196_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4768)
OSTM1 | 218196_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8667)
OSTM1 | 218196_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1913)
OSTM1 | 218196_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OSTM1 | 218196_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9635
p=0.5613
p=0.1574
p=0.9456
p=0.2669
p=0.2211
p=0.5247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2